

Factorize, Reconstruct, Enhance: A Unified Framework for Multimodal Sentiment Analysis

Zhilu Yang¹, Mingcheng Li^{2§}

¹Earth Science and Engineering, Imperial College London, London, United Kingdom

²College of Intelligent Robotics and Advanced Manufacturing, Fudan University

zy2425@ic.ac.uk

Abstract

Multimodal sentiment analysis (MSA) aims to predict human sentiment intensity and polarity by exploiting complementary information across modalities. This requires integrating heterogeneous signals while mitigating modality-specific noise and redundancy. Existing approaches rely on either coarse disentanglement strategies or insufficient cross-modal regularization, lacking explicit noise suppression and adaptive fusion mechanisms, which limits their ability to preserve fine-grained sentiment cues. In this work, we propose FUSE-Net, a Factorized and Unified Semantic Enhancement framework that decomposes each modality into shared, specific, and noise subspaces and applies contrastive learning, an information-gain constraint, and duality constraints for structured regularization. To preserve task-relevant semantics during factorization, we introduce a variational reconstruction pathway grounded in the information bottleneck principle. Furthermore, we design a multi-factor dynamic fusion module that adaptively aggregates modality subspaces from three complementary perspectives using a sample-conditioned scalar modulation score, factor-type coefficients, and branch-wise scalar attention, followed by an element-wise gate on the noise branch. Extensive experiments on three benchmark datasets (CMU-MOSI, CMU-MOSEI, and SIMSv2) demonstrate that FUSE-Net consistently outperforms strong baselines on key regression and ordinal classification metrics.

1. Introduction

Multimodal sentiment analysis (MSA) aims to infer human affective polarity by jointly modeling textual, acoustic, and visual signals, enabling a more comprehensive understanding of human emotions than unimodal approaches [23]. Over the past decade, existing research has largely evolved along two dominant paradigms. (i) *Fusion-based methods*

directly integrate multimodal features at either the feature or decision level, typically leveraging attention mechanisms or hierarchical aggregation to enhance cross-modal interaction and complementarity [25, 33, 44]. (ii) *Representation-learning approaches* aim to improve fusion quality by refining unimodal representations prior to interaction, often through redundancy reduction, modality decoupling, or self-supervised consistency objectives [28, 31, 42, 50]. Despite their success, these approaches generally rely on task-specific architectures and carefully engineered alignment objectives, limiting their generalizability and robustness under severe modality heterogeneity, redundancy, and noise contamination.

Recently, large language models (LLMs) and multimodal large language models (MLLMs) have emerged as a new paradigm for MSA by reformulating multimodal understanding as instruction-following or sequence generation tasks [1, 4, 20, 38, 41]. Through in-context learning and prompting, these models can exploit rich world knowledge and compositional reasoning to interpret implicit affective cues (e.g., sarcasm and commonsense context) while providing natural-language rationales that enhance interpretability [4, 38]. Representative works such as UniMSE unify multimodal inputs and outputs within an autoregressive framework [12], while LLaVAC adapts vision-language models for sentiment classification via instruction tuning [5, 20]. In addition, recent studies employ LLMs as auxiliary reasoning modules by verbalizing non-verbal signals into textual descriptions and performing reasoning over such representations [6, 16, 17, 32, 40]. Parameter-efficient adaptation techniques further enable scalable deployment by updating lightweight components while keeping large backbones frozen [11, 15]. Nevertheless, these approaches still face fundamental limitations. (i) Multimodal reasoning is implicitly embedded within highly entangled parameter spaces, lacking explicit controllability over fine-grained semantic factors, which hinders effective disentanglement of sentiment-relevant and nuisance information. (ii) Moreover, the absence of structured and sample-adaptive fusion mech-

[§]Corresponding author.

anisms limits reliable modulation of modality contributions, particularly under noisy or incomplete modality conditions.

To address these challenges, we propose FUSE-Net, a Factorized and Unified Semantic Enhancement Network for multimodal sentiment analysis. FUSE-Net is a structured representation-learning framework that explicitly disentangles multimodal features while maintaining semantic integrity and adaptive fusion capability. Specifically, (i) a hierarchical modality factorization module decomposes each modality into shared, specific, and noise subspaces, enabling explicit isolation of sentiment-consistent cues and nuisance variations; (ii) an information bottleneck-guided reconstruction channel preserves task-relevant semantics during factorization via variational reconstruction, preventing over-suppression of informative signals; and (iii) a multi-factor dynamic fusion module performs sample-adaptive integration through modulation scores, factor-type coefficients, and branch-wise attention, while further suppressing residual noise. Extensive experiments across multiple benchmarks demonstrate that FUSE-Net consistently outperforms strong baselines and achieves more robust and fine-grained sentiment estimation. “

2. Related Work

2.1. Multimodal Sentiment Analysis

Multimodal sentiment analysis (MSA) aims to infer affective states from heterogeneous signals across textual, acoustic, and visual modalities, where effective modeling requires both capturing cross-modal complementarity and mitigating modality-specific noise and redundancy. Existing studies primarily focus on improving representation quality and cross-modal interaction to address modality heterogeneity prior to fusion [12, 36, 39, 42]. Current approaches can be broadly categorized into two directions. Subspace decoupling methods decompose multimodal representations into shared and modality-specific components to disentangle complementary information from redundant or interfering signals while preserving modality-dependent sentiment cues. Representative works include DeCUR [36], which explicitly separates common and unique representations to reduce cross-modal redundancy and enhance complementary fusion, and DMRNet [39], which introduces probabilistic sampling strategies and embedding-blocking objectives to improve modality-wise disentanglement and the expressiveness of private subspaces. In parallel, self-supervised and auxiliary pre-training methods enhance cross-modal alignment without additional manual supervision by incorporating auxiliary objectives such as contrastive learning and masked modeling. For example, Self-MM [42] leverages cross-modal contrastive learning to align coherent multimodal representations while preserving modality-aware sentiment signals, whereas UniMSE [12]

adopts masked-modality prediction within a generative multitask framework to capture inter-modal dependencies and improve representation robustness. Despite these advances, existing methods generally rely on relatively coarse-grained decoupling or alignment strategies and lack explicit mechanisms for isolating sentiment-irrelevant noise factors, which may lead to contamination of learned representations and attenuation of subtle affective cues, especially under scenarios with low signal-to-noise ratio or unreliable modalities. Moreover, most approaches treat disentanglement and fusion as loosely coupled processes, without jointly optimizing semantic separation and downstream aggregation, which limits the overall controllability and robustness of multimodal reasoning. To address these limitations, FUSE-Net introduces hierarchical factorization with an explicit noise subspace, together with reconstruction-driven semantic regularization, enabling more precise disentanglement and more robust multimodal sentiment modeling.

2.2. Representation Learning for Multimodal Tasks

Representation learning provides a fundamental paradigm for modeling heterogeneous multimodal data by bridging modality gaps while suppressing irrelevant variations, and has been widely applied to tasks such as multimodal understanding, retrieval, and generation. Existing approaches can be broadly divided into two categories. Unified semantic space methods project heterogeneous modalities into a shared embedding space using large-scale pre-trained encoders or learnable modality projections, thereby facilitating direct cross-modal interaction and alignment at either the representation or token level. Representative works include MLA [50], BeamCLIP [13], and MMRL [8], which leverage large-scale pre-training and alignment objectives to improve semantic consistency across modalities. While effective, such approaches often rely on implicit alignment within a unified space, which may obscure modality-specific characteristics and limit fine-grained control over semantic components. In contrast, representation decoupling methods explicitly factorize multimodal representations into shared and private subspaces to preserve modality-specific information while reducing redundancy and semantic overlap. For instance, DCCL [37] employs disentangled contrastive learning to separate common semantics from modality-dependent features in video-text retrieval, while DisCo [34] further introduces a noise-related subspace to model spurious patterns and label corruption. Nevertheless, existing frameworks often suffer from two key limitations: they typically lack explicit modeling of noise-related factors, allowing nuisance variations to propagate into task-relevant representations, and they predominantly rely on geometric or contrastive objectives without incorporating task-guided semantic preservation

mechanisms such as reconstruction-based regularization, which limits the stability, interpretability, and controllability of learned latent factors. Furthermore, the interaction between disentangled subspaces and downstream fusion is often underexplored, leading to suboptimal utilization of structured representations. In contrast, FUSE-Net combines hierarchical factorization with an explicit noise branch and integrates information bottleneck-driven reconstruction, enabling cleaner subspace separation while preserving sentiment-relevant semantics and facilitating more effective downstream fusion.

3. Methodology

3.1. Task Definition

Given a multimodal video segment consisting of three modalities, we represent it as $U = (\mathbf{X}_L, \mathbf{X}_V, \mathbf{X}_A)$, where $\mathbf{X}_L \in \mathbb{R}^{T_L \times d_L}$, $\mathbf{X}_A \in \mathbb{R}^{T_A \times d_A}$, and $\mathbf{X}_V \in \mathbb{R}^{T_V \times d_V}$ correspond to the language, audio, and visual modalities, respectively. For each modality $m \in \{L, V, A\}$, the feature sequence is $\mathbf{X}_m \in \mathbb{R}^{T_m \times d_m}$, where T_m and d_m denote the sequence length and feature dimension. Our goal is to recognize the utterance-level sentiments by utilizing the multimodal data.

3.2. Overall Framework

Figure 1 illustrates the overall workflow of FUSE-Net. Given raw feature sequences from the three modalities, the framework proceeds as follows. (i) The language, visual, and audio features are first fed into modality-specific encoders to obtain contextualized representations $(\mathbf{H}_L, \mathbf{H}_V, \mathbf{H}_A)$. (ii) These representations are then sent to the Hierarchical Modality Factorization (HMF) module, which decomposes each modality into shared, specific, and noise subspaces to separate sentiment-relevant information from nuisance factors. (iii) The decomposed factors are further regularized by the Modality Reconstruction Channel (MRC), which preserves task-relevant semantics through variational reconstruction while suppressing redundancy. (iv) The refined factors are then integrated by the Multi-factor Dynamic Fusion (MDF) module, where sample-conditioned modulation, factor-type coefficients, and branch-wise attention jointly determine the contribution of each branch, followed by noise suppression through an element-wise gate. (v) Finally, the fused representation $\mathbf{H}$ fused is fed into a prediction head to produce the utterance-level sentiment estimate $\hat{y}$. Subsequent sections detail the HMF, MRC, and MDF modules.

3.3. Hierarchical Modality Factorization (HMF)

Direct cross-modal interaction over raw unimodal representations is often suboptimal for multimodal sentiment analysis (MSA), since each modality generally entangles three

heterogeneous sources of information, namely modality-invariant sentiment semantics, modality-specific affective cues, and task-irrelevant nuisance variations. Without explicit factorization, redundant shared information may be repeatedly aggregated across modalities, distinctive affective evidence may be suppressed by dominant signals, and irrelevant variations may be unintentionally introduced into the fusion process. Existing shared-private decoupling methods [10, 36, 39] partially mitigate this issue by separating shared and non-shared components, yet they still rely on a relatively coarse decomposition that does not explicitly isolate sentiment-irrelevant factors. As a consequence, genuine modality-specific cues and nuisance variations may remain entangled within the same branch, leading to nuisance leakage and reduced robustness in downstream fusion. To address this issue, we introduce Hierarchical Modality Factorization (HMF), which decomposes each modality into three complementary subspaces: a shared subspace for modality-invariant sentiment semantics, a specific subspace for modality-exclusive but sentiment-relevant cues, and a noise subspace for sentiment-irrelevant variations. Such a tri-factor decomposition provides a cleaner representational basis for subsequent reconstruction and adaptive fusion.

Formally, given the representation $\mathbf{H}_m$ of modality m , HMF decomposes it into three components: the shared representation $\mathbf{H}_m^h$, the specific representation $\mathbf{H}_m^s$, and the noise representation $\mathbf{H}_m^n$. Compared with conventional shared-private factorization, the explicit introduction of $\mathbf{H}_m^n$ prevents nuisance information from being implicitly absorbed into the private branch, thereby encouraging sentiment-related knowledge to be concentrated in $(\mathbf{H}_m^h, \mathbf{H}_m^s)$ and yielding more controllable factors for downstream interaction. Specifically, HMF is learned under three cooperative objectives. To align modality-invariant semantics while preserving the distinction between shared and specific factors, we introduce a contrastive separation objective. Let $\Omega = \{L, V, A\}$ denote the modality set, and define

$$\phi(u, v) = \exp\left(\frac{\text{sim}(u, v)}{\tau}\right), \quad (1)$$

where $\text{sim}(\cdot, \cdot)$ denotes cosine similarity and τ is a temperature hyperparameter. The resulting objective is given by

$$\mathcal{L}_{\text{contrast}} = -\frac{1}{6} \sum_{a \in \Omega} \sum_{\substack{b \in \Omega \\ b \neq a}} \log \left(\frac{\phi(\mathbf{H}_a^h, \mathbf{H}_b^h)}{\phi(\mathbf{H}_a^h, \mathbf{H}_b^h) + \phi(\mathbf{H}_a^h, \mathbf{H}_b^s)} \right). \quad (2)$$

This term promotes cross-modal agreement in the shared semantic space while preventing modality-specific affective information from collapsing into the shared branch.

To regulate the allocation of sentiment-related information across different branches, we further impose an information-gain constraint by attaching auxiliary sentiment estimators to $\mathbf{H}_m^h$, $\mathbf{H}_m^s$, and $\mathbf{H}_m^n$. This objective en-

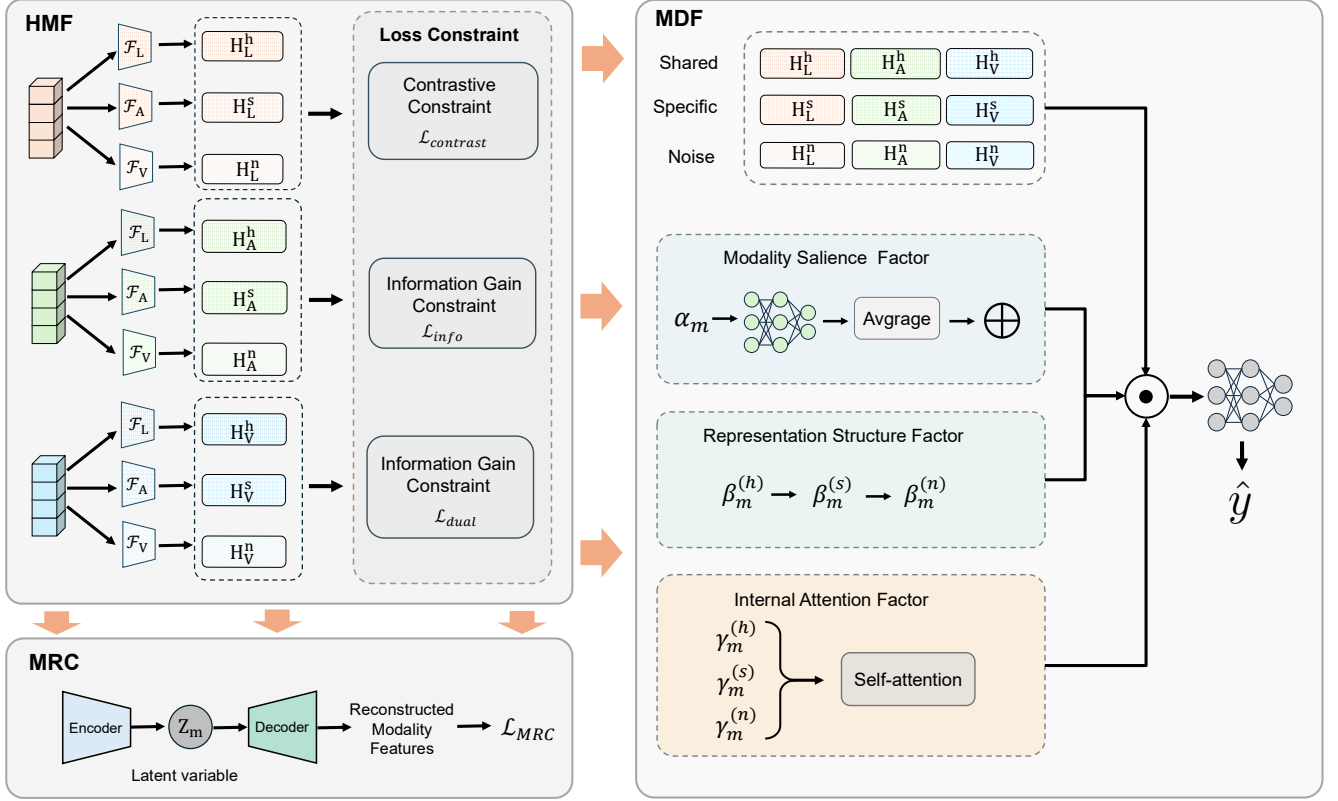


Figure 1. Overview of FUSE-Net. Multimodal inputs are first processed by the HMF module for hierarchical fusion across modalities. The MRC module captures cross-modal interactions to exploit complementary information, while the MDF module refines the fused representations to improve robustness and discriminability. The resulting features are fed into the prediction head for final inference.

courages sentiment-discriminative information to accumulate in the shared and specific branches while suppressing such information in the noise branch:

$$\mathcal{L}_{info} = \frac{1}{|\Omega|} \sum_{m \in \Omega} (-\ell_h^m - \ell_s^m + \ell_n^m), \quad (3)$$

where ℓ_h^m , ℓ_s^m , and ℓ_n^m denote branch-specific information scores produced by the corresponding auxiliary predictors.

To preserve structural compatibility between the shared and specific subspaces, we additionally employ a dual consistency regularizer. Specifically, mutual recoverability is enforced through a pair of mappings g_m and g_m^{-1} , which stabilizes the decomposition process and reduces representation drift:

$$\mathcal{L}_{dual} = \frac{1}{|\Omega|} \sum_{m \in \Omega} \left(\left\| \mathbf{H}_m^h - g_m(\mathbf{H}_m^s) \right\|_2^2 + \left\| \mathbf{H}_m^s - g_m^{-1}(\mathbf{H}_m^h) \right\|_2^2 \right). \quad (4)$$

By combining the above objectives, the overall optimization target of HMF can be written as

$$\mathcal{L}_{HMF} = \lambda_{contrast} \mathcal{L}_{contrast} + \lambda_{info} \mathcal{L}_{info} + \lambda_{dual} \mathcal{L}_{dual}, \quad (5)$$

where $\lambda_{contrast}$, λ_{info} , and λ_{dual} are trade-off coefficients that balance semantic alignment, branch-wise information

allocation, and structural consistency across subspaces, respectively.

3.4. Modality Semantic Reconstruction

A central challenge in multimodal sentiment analysis lies in maintaining semantic sufficiency under heterogeneous and noisy observations. In practice, affective cues are often subtle, context-sensitive, and sparsely distributed, whereas modality-internal variations may exhibit high variance and easily dominate the representation space [3, 45]. Although HMF explicitly decomposes each modality into shared, specific, and noise components, the resulting factors may still suffer from semantic incompleteness when the input data are affected by low signal-to-noise ratios, partial corruption, or severe modality imbalance, thereby weakening downstream fusion and increasing the risk of shortcut correlations [9]. More specifically, factorization-oriented objectives mainly regulate the relative geometry among latent components, but they do not explicitly ensure that the retained information remains sufficient to recover affect-relevant modality semantics. Under such circumstances, overly aggressive decomposition may suppress weak but crucial emotional evidence, and the resulting semantic defi-

ciency can propagate to the fusion stage, leading to unstable predictions. Therefore, in addition to factor separation, an explicit semantic preservation mechanism is required to stabilize latent factor learning while continuing to suppress redundant and nuisance information.

To this end, we introduce the Modality Reconstruction Channel (MRC), an information bottleneck-guided reconstruction regularizer derived from the Variational Information Bottleneck (VIB) principle [2]. The underlying objective is to enforce a sufficiency–minimality trade-off: the latent representation should preserve enough information to recover modality semantics relevant to sentiment understanding, while remaining compact enough to filter out residual redundancy and nuisance disturbances, thereby realizing denoising through controlled compression [14, 26]. Concretely, for each modality m , the three factors produced by HMF, namely the shared factor H_m^h , the specific factor H_m^s , and the noise factor H_m^n , are concatenated into a unified representation:

$$\mathbf{H}_m^{\text{concat}} = [\mathbf{H}_m^h; \mathbf{H}_m^s; \mathbf{H}_m^n]. \quad (6)$$

The concatenated representation is then passed into a modality-specific variational encoder $\mathcal{E}_\phi$, which parameterizes a diagonal Gaussian posterior $q_\phi(z_m | \mathbf{H}_m^{\text{concat}})$. A latent variable z_m is sampled via the reparameterization trick and subsequently fed into a decoder $\mathcal{D}_\theta$ to reconstruct the modality representation before factorization, yielding $\hat{\mathbf{H}}_m$. Accordingly, the MRC objective is formulated as

$$\mathcal{L}_{\text{MRC}} = \sum_{m \in \Omega} \left(\|\hat{\mathbf{H}}_m - \mathbf{H}_m\|_2^2 + \beta D_{\text{KL}}(q_\phi(z_m | \mathbf{H}_m^{\text{concat}}) \| \mathcal{N}(0, I)) \right). \quad (7)$$

In this objective, the reconstruction term preserves semantic sufficiency by penalizing the loss of modality content that contributes to sentiment reasoning, whereas the KL regularization constrains the information capacity of the latent channel so as to suppress residual redundancy and nuisance variations; the coefficient β controls the balance between these two effects. Notably, applying MRC to the concatenation of all factor branches allows the reconstruction signal to jointly refine the shared, specific, and noise subspaces. This design mitigates the risk of over-purification introduced by decomposition alone and produces semantically more robust factors for the subsequent MDF module.

3.5. Multi-factor Dynamic Fusion

Multimodal fusion in MSA remains difficult even with structured factorization and reconstruction-based refinement, because the task exhibits variability at multiple levels [3]. First, the effective contribution of each modality is highly sample-dependent: acoustic and visual streams can be corrupted by background noise, occlusions, or recording artifacts, and different utterances may rely on different modalities to express sentiment [28, 46]. Second, the relative importance of shared versus modality-specific cues is

context-dependent, and residual nuisance information may still be unevenly distributed across factor branches, motivating dynamic and fine-grained fusion rather than static weighting [19, 33]. Under these conditions, static fusion strategies such as direct concatenation or fixed weighting are prone to suboptimal information allocation, leading to low utilization of complementary cues and unstable predictions when noisy or misleading components dominate. Therefore, an effective fusion mechanism should adapt across modalities and factor types, while further suppressing residual nuisance components, rather than relying on a single-level gating rule.

To address these issues, we propose a Multi-factor Dynamic Fusion (MDF) module that performs dynamic fusion over the factors produced by HMF (and stabilized by MRC). MDF decomposes fusion control into three complementary factors operating at different scopes, enabling sample-wise calibration of branch allocation and factor contribution. Specifically, a sample-conditioned scalar modulation score α_m is computed from the concatenation of the shared and specific factors of modality m ; the structural prior $\beta^{(x)}$ is a learnable scalar coefficient for each factor type $x \in \{h, s, n\}$; and $\gamma_m^{(x)}$ is a branch-wise scalar attention score estimated from each modality–factor representation. Their product forms an unnormalized logit for each modality–factor branch. Formally, the sample-conditioned scalar modulation score is computed as

$$\alpha_m = \text{MLP}([\mathbf{H}_m^h; \mathbf{H}_m^s]), \quad (8)$$

the branch-wise scalar score is

$$\gamma_m^{(x)} = \text{Attn}(\mathbf{H}_m^{(x)}), \quad x \in \{h, s, n\}, \quad (9)$$

and the learnable factor-type coefficient is denoted by $\beta^{(x)}$. The unnormalized logit for each modality–factor branch is then

$$\ell_m^{(x)} = \alpha_m \cdot \beta^{(x)} \cdot \gamma_m^{(x)}, \quad x \in \{h, s, n\}. \quad (10)$$

We normalize the three branch logits within each modality:

$$w_m^{(x)} = \frac{\exp(\ell_m^{(x)})}{\sum_{x' \in \{h, s, n\}} \exp(\ell_m^{(x')})}, \quad x \in \{h, s, n\}. \quad (11)$$

This within-modality normalization explicitly redistributes each modality’s contribution among shared, specific, and noise factors on a per-sample basis, improving the controllability of information allocation before cross-modal aggregation. The weighted aggregation across modalities yields

$$\mathbf{F}^{(s)} = \sum_{m \in \Omega} w_m^{(s)} \mathbf{H}_m^{(s)}, \quad (12)$$

$$\mathbf{F}^{(h)} = \sum_{m \in \Omega} w_m^{(h)} \mathbf{H}_m^{(h)}, \quad (13)$$

$$F^{(n)} = \sum_{m \in \Omega} w_m^{(n)} \cdot \text{Gate}(\mathbf{H}_m^{(n)}), \quad (14)$$

where $\text{Gate}(\mathbf{h}) = \mathbf{h} \odot \sigma(\mathbf{W}_g \mathbf{h} + \mathbf{b}_g)$ further suppresses residual noise via learnable sigmoid modulation. Finally, the three fused factors are concatenated and passed into a two-layer MLP to produce the sentiment prediction:

$$\mathbf{H}_{\text{fused}} = \text{Concat}[F^{(h)}, F^{(s)}, F^{(n)}], \quad (15)$$

$$\hat{y} = \text{MLP}(\mathbf{H}_{\text{fused}}). \quad (16)$$

The overall training objective combines the task loss with the regularization losses induced by HMF and MRC:

$$\mathcal{L}_{\text{total}} = \lambda_{\text{task}} \mathcal{L}_{\text{task}} + \mathcal{L}_{\text{HMF}} + \lambda_{\text{MRC}} \mathcal{L}_{\text{MRC}}, \quad (17)$$

where $\mathcal{L}_{\text{task}}$ denotes mean squared error on the continuous sentiment label, and λ terms balance the contributions of the individual components. An optional siamese-style noise regularizer is implemented in code but omitted here since its coefficient is set to zero in all experiments.

4. Experiments

4.1. Datasets and Evaluation Metrics

We evaluate FUSE-Net on three public benchmarks, CMU-MOSI [43], CMU-MOSEI [45], and SIMSv2 [22]. For CMU-MOSI and CMU-MOSEI, we follow the standard word-aligned setting in MSA, where acoustic and visual streams are aligned to word timestamps via interval aggregation. For SIMSv2, we use the original temporal sequences provided by the benchmark without applying additional cross-modal alignment. CMU-MOSI contains 2,199 utterances and is split into 1,284 training, 229 validation, and 686 test samples. CMU-MOSEI contains 22,856 utterances and is split into 16,326 training, 1,871 validation, and 4,659 test samples. Both datasets provide real-valued sentiment intensity labels in the range $[-3, 3]$. SIMSv2 is a Chinese multimodal sentiment benchmark with regression targets in $[-1, 1]$. For MOSI and MOSEI, we report MAE, Pearson correlation (Corr), Acc_7 , and binary Acc_2 and F1 under two standard label mappings. For SIMSv2, we report MAE, Corr, Acc_2 , Acc_3 , Acc_5 , and F1 under the commonly used discretization protocol.

4.2. Implementation Details

Feature Extraction. Text is encoded using the RoBERTa model [21], where contextualized token embeddings are obtained from the last Transformer layer. We apply masked mean pooling over valid tokens to derive a fixed-length utterance-level representation. For the acoustic modality, we adopt COVAREP descriptors [7], which capture low-level prosodic and spectral features such as pitch, energy, and voice quality. For the visual modality, we use Facet facial features [29], which provide frame-level facial action

units and expression-related cues. All modality features are temporally aligned at the word level following standard pre-processing protocols.

Experimental Setup. All methods are implemented in PyTorch and trained on a single NVIDIA RTX 4090. Visual and acoustic streams are encoded by a one-layer BiGRU with hidden size 128, and projected to 128 dimensions. Text is encoded by the BERT-base model and projected to the same dimensionality. We optimize with AdamW and use ReduceLROnPlateau with validation-based early stopping. Results are averaged over five random seeds.

4.3. Comparison with State-of-the-Art Methods

Table 1 and Table 2 present the comparisons between FUSE-Net and a series of representative as well as reproducible baselines, including fusion-oriented methods such as TFN [44], LMF [23], ICCN [30], and MulT [33], together with disentanglement- and regularization-based approaches, including MISA [10], Self-MM [42], MMIM [9], MSG [19], PriSA [24], DMD [18], MIM [47], DTN [48], and ALMT [49]. Overall, FUSE-Net achieves the lowest MAE on all three benchmarks, namely MOSI, MOSEI, and SIMSv2, while also obtaining the best Acc_7 on both MOSI and MOSEI, demonstrating its superiority in terms of both regression accuracy and fine-grained sentiment ranking.

To further examine the empirical advantage of the proposed framework, we take DTN [48], a strong and competitive disentanglement-based baseline, as a reference and compute relative improvements on directly comparable metrics. The results can be interpreted from the following aspects. (i) On MOSI, FUSE-Net improves MAE, Corr, and Acc_7 over DTN with an average relative gain of 2.88%, among which the largest improvement is achieved on MAE at 3.91%. On MOSEI, the same set of metrics exhibits an even larger average relative gain of 4.21%, with the most pronounced improvement again appearing on MAE at 7.87%. These results indicate that the proposed framework is particularly effective in reducing estimation error for continuous sentiment prediction, which can be attributed to its stronger ability to isolate nuisance variations and preserve sentiment-relevant information during representation learning. (ii) For the binary classification setting on MOSI and MOSEI, although DTN does not report the corresponding Acc_2 and F1 results, FUSE-Net consistently achieves the best performance under both standard label mappings among all methods with available results. This observation suggests that the advantages of FUSE-Net are not limited to regression-oriented metrics, but also transfer effectively to decision-oriented sentiment prediction scenarios. (iii) On SIMSv2, FUSE-Net outperforms DTN on MAE, Corr, Acc_2 , Acc_3 , and Acc_5 , yielding an average relative gain of 1.96% across these comparable metrics, with the

Methods	MOSI Dataset					MOSEI Dataset				
	MAE ↓	Corr ↑	Acc ₇ ↑	Acc ₂ ↑	F1 ↑	MAE ↓	Corr ↑	Acc ₇ ↑	Acc ₂ ↑	F1 ↑
TFN [44] [†]	0.947	0.673	34.46	77.99/79.08	77.95/79.11	0.572	0.714	51.60	78.50/81.89	78.96/81.74
LMF [23] [†]	0.950	0.651	33.82	77.90/79.18	77.80/79.15	0.576	0.717	51.59	80.54/83.48	80.94/83.36
ICCN [30]	0.862	0.714	39.0	-/83.0	-/83.0	0.565	0.713	51.6	-/84.2	-/84.2
MuT [33] [†]	0.879	0.702	36.91	79.71/80.98	79.63/80.95	0.559	0.733	52.84	81.15/84.63	81.56/84.52
MISA [10] [†]	0.776	0.778	41.37	81.84/83.54	81.82/83.58	0.568	0.724	-	82.59/84.23	82.67/83.97
Self-MM [42] [†]	0.708	0.796	46.67	83.44/85.46	83.36/85.43	0.531	0.764	53.87	83.76/85.15	83.82/84.90
MMIM [9]*	0.718	0.797	46.64	83.38/85.82	83.29/85.81	0.537	0.759	53.42	82.08/85.14	82.51/85.11
MSG [19]	0.748	0.782	47.3	-/85.6	-/85.6	0.583	0.787	52.8	-/85.4	-/85.4
PriSA [24]*	0.719	0.782	47.1	83.3/85.4	83.1/85.4	0.536	0.761	53.95	82.1/85.2	83.3/85.2
DMD [18]*	-	-	43.9	-/84.9	-/85.0	-	-	53.1	-/85.2	-/85.2
MIM [47]*	0.718	0.792	46.4	-/84.8	-/84.8	0.579	0.779	51.8	-/85.7	-/85.6
DTN [48]*	0.716	0.790	47.5	-/85.1	-/85.1	0.572	0.765	52.3	-/85.5	-/85.5
FUSE-Net (Ours)	0.688	0.798	49.27	83.53/85.83	83.75/85.96	0.527	0.772	54.32	84.52/85.88	84.5/86.14

Table 1. Comparison results on the MOSI and MOSEI datasets. [†]: results from [27]; *: results reproduced from the open-source codebase with hyperparameters provided in original papers. For Acc₂ and F1, we report two settings: non-negative/negative (left) and positive/negative (right). Bold indicates the best results.

Table 2. Comparison on SIMSv2. Notes: [†] from [27]; * reproduced with authors’ hyperparameters. Best values are in bold.

Methods	SIMSv2 Dataset					
	MAE ↓	Corr ↑	Acc ₂ ↑	Acc ₃ ↑	Acc ₅ ↑	F1 ↑
TFN [44] [†]	0.329	0.640	77.95	70.21	51.93	77.74
LMF [23] [†]	0.367	0.557	74.18	64.90	47.79	73.88
MuT [33]*	0.304	0.705	79.30	72.63	53.29	79.43
Self-MM [42] [†]	0.322	0.678	79.11	72.34	53.00	79.05
CENet [35] [†]	0.310	0.699	79.56	73.10	53.04	79.63
ALMT [49]*	0.308	0.700	79.59	71.86	52.90	79.51
DMD [18]*	0.305	0.702	78.87	72.01	53.18	79.21
MIM [47]*	0.310	0.694	77.56	71.45	52.87	78.56
DTN [48]*	0.302	0.701	78.29	72.56	53.71	78.12
FUSE-Net (Ours)	0.297	0.712	79.59	73.69	55.51	79.53

largest gain of 3.35% observed on Acc₅. Such results indicate that the proposed framework maintains stronger robustness when sentiment labels are evaluated under finer-grained discretization protocols.

4.4. Ablation Studies

Table 3 summarizes the ablation results on the MOSI dataset, where all variants are evaluated over five random

Table 3. Ablation results on the MOSI dataset.

Models	Metrics				
	MAE ↓	Corr ↑	Acc ₇ ↑	Acc ₂ ↑	F1 ↑
FUSE-Net (Full)	0.688	0.798	49.27	83.53/85.83	83.75/85.96
Importance of Modalities					
w/o Language	1.246	0.302	23.5	51.6/53.4	51.7/53.6
w/o Audio	0.758	0.796	45.7	82.4/84.6	82.6/84.7
w/o Visual	0.723	0.801	46.4	83.5/85.2	83.7/85.3
Importance of Proposed Components					
w/o HMF	0.754	0.762	43.0	80.2/82.4	80.4/82.5
w/o MRC	0.729	0.785	45.7	82.4/84.7	82.5/84.8
w/o MDF	0.712	0.796	46.3	82.9/85.1	83.0/85.3

seeds. To quantify the contribution of each component more clearly, we report relative changes by treating each ablated variant as the reference model. Specifically, for MAE, we report the relative reduction achieved by the full model with respect to the corresponding ablated counterpart, whereas for Corr, Acc₇, Acc₂, and F1, we report the relative improvement of the full model over the same reference.

Importance of Proposed Components. The component-wise ablation results provide further insight into the role of

Table 4. Case study on two representative utterances from a randomly sampled subset of the MOSI test set.

Case	Transcript	GT	DTN	MISA	FUSE-Net
1	I was like you know they're both hasbro toys they're both movies that are made for children	neu	pos (0.77)	pos (0.66)	neu (0.14)
2	There's a kevin costner prince of thieves version which was supposed to be the dark and edgy version of his time i mean but that was what thirteen years ago fourteen years ago probably more than that	neg	neu (-0.43)	neu (-0.46)	neg (-0.56)

each proposed module. (i) Contribution of HMF. Among all components, HMF accounts for the largest performance gain. Removing HMF leads to the most substantial degradation, with MAE increasing to 0.754 and Acc_7 dropping to 43.0. This observation highlights the importance of explicitly disentangling shared, specific, and noise-related factors, which effectively reduces nuisance leakage into sentiment-bearing representations. (ii) Contribution of MRC. The removal of MRC results in a consistent deterioration across all evaluation metrics, indicating that the reconstruction-based regularization plays a complementary role in stabilizing latent factor learning. In particular, MRC helps preserve task-relevant semantics during decomposition and alleviates the risk of over-purification caused by aggressive factorization. (iii) Contribution of MDF. Excluding MDF also leads to a noticeable, though relatively smaller, performance decline. The same tendency is reflected in the binary evaluation results under both label mappings, suggesting that sample-adaptive factor reweighting further improves robustness beyond static fusion strategies.

Importance of Modalities. The modality ablation results reveal a clear hierarchy in modality contribution on MOSI. (i) Role of language. Language serves as the dominant source of sentiment information. When the textual modality is removed, performance degrades dramatically, with MAE rising to 1.246 and Corr dropping to 0.302. This result confirms that lexical semantics constitute the primary basis for reliable sentiment inference in this benchmark. (ii) Role of audio. Audio provides stable complementary information. Removing the acoustic modality causes a clear decline across both regression and ordinal evaluation metrics, suggesting that prosodic patterns offer useful cues for resolving sentiment intensity and pragmatic nuances when textual evidence alone is insufficient. (iii) Role of vision. Visual information also contributes positively, although its effect is relatively weaker in the current setting. After removing the visual modality, MAE, Acc_7 , Acc_2 , and F1 all decrease, whereas Corr changes only marginally. This trend is consistent with the nature of facial cues, which are informative but often sparse and noisy, making them more vulnerable to speaker variation and recording artifacts.

4.5. Case Study

To provide a more fine-grained understanding of model behavior, Table 4 reports two representative utterances selected from the same sampled subset. In Case 1, the ground-truth sentiment is neutral, whereas DTN and MISA both overestimate the utterance as positive, yielding prediction scores of 0.77 and 0.66, respectively. In contrast, FUSE-Net predicts a score of 0.14, which remains within the correct neutral interval. In Case 2, the utterance conveys mild negative sentiment. DTN and MISA produce scores of -0.43 and -0.46, respectively, both of which are assigned to the neutral category under the three-way discretization criterion. By comparison, FUSE-Net outputs -0.56 and correctly identifies the utterance as negative. This advantage can be attributed to the explicit introduction of the noise subspace (H_m^n), which absorbs sentiment-irrelevant disturbances such as background noise or exaggerated paralinguistic variations, thereby allowing the specific subspace (H_m^s) to focus more effectively on sentiment-bearing affective cues. Overall, these examples further support the central design motivation of FUSE-Net: explicit nuisance separation together with semantic preservation can reduce prediction ambiguity and improve robustness on utterances with subtle or weak sentiment expressions.

5. Conclusion

In this paper, we present FUSE-Net, a factorized and unified framework for multimodal sentiment analysis. Concretely, we propose a hierarchical modality factorization (HMF) module that decomposes each modality into shared, specific, and noise subspaces to separate sentiment-relevant information from nuisance variations. Additionally, we introduce an information bottleneck-guided reconstruction channel (MRC) to preserve task-relevant semantics and stabilize factor learning. Furthermore, a multi-factor dynamic fusion (MDF) module adaptively reweights modality contributions for robust multimodal integration. Extensive experiments on MOSI, MOSEI, and SIMSv2 demonstrate consistent improvements over strong baselines, and ablation studies validate the effectiveness of each component.

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